

ResolveIQ

AI-Powered Frictionless Dispute & Chargeback Resolution

Project Description | Round 1 Submission | 2025

Challenge Track: Frictionless Dispute & Chargeback Resolution

1. Executive Summary

Every year, card issuers process tens of millions of dispute and chargeback cases. Today, a resolution that should take minutes takes 14 to 45 days — frustrating card members, straining merchant relationships, and costing financial institutions hundreds of millions of dollars in operational overhead.

"ResolveIQ cuts the dispute cycle from weeks to under 5 minutes for 70% of cases — with AI-powered evidence collection, a dual-fairness weighing model, and plain-English decision explanations that both card members and merchants can trust."

ResolveIQ is an end-to-end AI dispute resolution platform built specifically for card issuers like American Express. It auto-collects transaction evidence, fairly weighs both the card member and merchant perspective, and delivers a transparent, explainable outcome — with zero manual intervention for the majority of cases.

< 5 min Resolution time (auto cases)	70% Cases auto-resolvable	\$40+ Saved per dispute	2× NPS uplift potential
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2. Problem Statement

2.1 The Dispute Experience is Broken

When a card member disputes a charge today, both sides enter a slow, opaque, and frustrating process. The card member must manually gather receipts and written statements. The merchant must respond to blanket requests within tight deadlines. The issuer's team must review evidence manually, referencing chargeback reason codes, and make a judgment call — often without full context. This process regularly takes 2 to 6 weeks.

2.2 Pain Points by Stakeholder

Stakeholder	Primary Pain	Business Cost
Card Members	No visibility into status	Drop disputes early due to friction
Merchants	Blanket evidence requests	No insight into decision rationale
Card Issuers	High cost per case (\$25–\$50)	Agent bandwidth wasted on simple cases
Regulators	Need audit trails	Manual processes are hard to audit at scale

2.3 Why Now

- ▶ Large language models and NLP can now parse unstructured evidence at scale
- ▶ Card members expect real-time digital experiences — 14-day cycles are unacceptable

- ▶ Regulatory pressure (CFPB, EU AI Act) is increasing focus on explainable financial decisions
- ▶ Merchant dissatisfaction with chargeback processes is a growing competitive risk for card networks

3. Solution Overview

ResolveIQ operates in three intelligent layers that work together to deliver fast, fair, and transparent dispute resolution.

Layer 1 — Evidence Autopilot (Zero-Touch Evidence Collection)

The moment a dispute is filed, Evidence Autopilot automatically gathers all relevant data from multiple sources — without requiring manual uploads from either party as a starting point.

Sources auto-collected:

- ▶ **Transaction metadata:** Timestamp, merchant category code (MCC), amount, terminal ID, card presence flag
- ▶ **Merchant data:** Delivery confirmations, signed receipts, return policy text via merchant API webhooks
- ▶ **Card member history:** Relationship tenure, prior dispute frequency, spending pattern baseline
- ▶ **Email parsing:** Purchase confirmations and delivery notifications parsed via NLP (spaCy + Hugging Face DistilBERT)
- ▶ **AMEX chargeback codes:** Automatic mapping of the disputed transaction to the most relevant chargeback reason code(s)
- ▶ **Case similarity index:** Retrieval-augmented lookup of past resolved disputes with similar signatures

Both parties can supplement this auto-collected evidence, but the baseline is never zero — reducing abandonment and incomplete submissions dramatically.

Layer 2 — FairScore™ Weighing Model (Dual-Perspective AI)

FairScore™ is the decision engine at the heart of ResolveIQ. It evaluates evidence from both sides using a transparent, configurable scoring model — not a black-box classifier.

FairScore™ evaluates six dimensions:

- ▶ **Evidence strength:** Quality and completeness of documentation from each side (receipts, signatures, timestamps)
- ▶ **Reason code alignment:** How well the dispute claim matches the cited chargeback category's evidentiary requirements
- ▶ **Merchant history:** Merchant's historical dispute win/loss rate, flags serial-dispute merchants
- ▶ **Card member history:** Card member's dispute frequency, prevents bad-faith claim patterns
- ▶ **Case precedent:** Similarity score to past resolved cases, retrieved via vector search (pgvector)
- ▶ **Policy compliance:** Whether the transaction falls within policy windows and amount thresholds

Output: a confidence score (0–100), a recommended outcome (Favour Card Member / Favour Merchant / Escalate to Human Review), and a structured evidence summary passed to Layer 3.

Layer 3 — Explainability Engine ("Why was this decided?")

Every ResolveIQ decision is accompanied by a plain-English explanation generated by an LLM, grounded in the structured FairScore™ output. This is not a post-hoc justification — it is generated directly from the evidence weights and precedents used in the decision.

Example explanation for a card member:

"We resolved this dispute in your favour because: the merchant was unable to provide a signed delivery confirmation within the required window, the transaction amount did not match the amount on your receipt, and 94% of similar cases in our system were resolved in the card member's favour. This decision was made in accordance with AMEX Chargeback Code 4513."

Example explanation for a merchant:

"We were unable to resolve this dispute in your favour because: the delivery confirmation provided post-dates the card member's dispute filing, and the item description on the receipt does not match the disputed transaction. We recommend reviewing your order fulfilment timestamps."

This transparency layer dramatically reduces appeal rates, builds trust with both parties, and provides a defensible audit trail for compliance purposes.

4. Innovation & Differentiation

What Makes ResolveIQ Different

Innovation	Why It Matters
Zero-Touch Evidence Collection	Most solutions ask users to upload documents manually. ResolveIQ pulls evidence automatically from connected sources, making the starting point never empty. This is the single biggest friction reducer in the dispute funnel.
Dual-Perspective Fairness Model	Most dispute tools are built to protect the card member by default. ResolveIQ explicitly models both sides. This is novel — and critical to merchant satisfaction, which is a real competitive concern for card networks.
Explainable AI by Design	Explainability is not bolted on as a summary. The reasoning is structural: every factor in FairScore™ is directly reflected in the explanation. This maps to regulatory trends (EU AI Act Article 22, CFPB) and reduces appeal rates.
RAG-Based Precedent Engine	Using past resolved disputes as a knowledge base ensures decisions are consistent over time and improvable through human feedback. No other dispute tool treats precedent as a first-class input.
Immutable Audit Trail	Every decision, evidence item, and action is logged in an append-only, hash-chained audit store — producing a verifiable record that satisfies both internal compliance and external regulatory requirements.

5. Technical Architecture

5.1 System Components

Component	Description
Card Member Portal	React + Tailwind. One-click dispute filing, real-time status tracker (WebSocket), evidence supplement upload, plain-language decision view.
Merchant Portal	React + Tailwind. Automated dispute notification, evidence submission interface, deadline countdown, decision rationale view.
AMEX Ops Dashboard	React + Recharts. Human review queue (low-confidence cases), bulk audit log, model performance monitoring.
Evidence Autopilot Service	Python + FastAPI. Orchestrates data collection from transaction DB, merchant webhooks, email NLP parser, and AMEX chargeback code mapper.
NLP Evidence Parser	spaCy + Hugging Face DistilBERT. Extracts entities (dates, amounts, merchant names, item descriptions) from unstructured email and document text.
FairScore™ Engine	XGBoost classifier + SHAP values. Trained on historical dispute outcomes. Produces confidence scores and per-feature importance weights.
Precedent Retrieval (RAG)	pgvector on PostgreSQL. Stores embeddings of past resolved cases. Retrieved by FairScore™ to inform decisions and by Explainability Engine to cite precedent.
Explainability Engine	LLM (Claude / GPT-4) with structured prompt grounded in FairScore™ output. Generates card member and merchant-facing decision explanations.
Audit Log Service	Append-only PostgreSQL with SHA-256 hash chaining per record. Every action, decision, and API call is logged immutably.
Notification Service	AWS SNS + SES. Real-time push and email notifications at every status change for both parties.

5.2 Tech Stack

Layer	Technology	Purpose
Frontend	React, Tailwind CSS, WebSockets	Card member, merchant, and ops portals
Backend / API	Python (FastAPI), Node.js	Core services and orchestration
NLP / AI	spaCy, Hugging Face DistilBERT	Evidence parsing from emails and documents
Decision Model	XGBoost + SHAP	FairScore™ weighing engine with explainability
Precedent Retrieval	pgvector, PostgreSQL	RAG-based case similarity engine
LLM Layer	Claude API / GPT-4	Plain-English decision explanation generation
Cloud	AWS (Lambda, S3, RDS, SNS, SES)	Serverless compute, storage, notifications

Layer	Technology	Purpose
Audit Store	Append-only PostgreSQL	Hash-chained immutable decision log
Monitoring	Prometheus + Grafana	Model drift, latency, and fairness metrics

5.3 Data Flow

- ▶ Card member files dispute → Evidence Autopilot triggers automatically
- ▶ Autopilot collects: transaction data, merchant webhook response, NLP-parsed email evidence
- ▶ FairScore™ runs: scores evidence strength, reason code alignment, history, precedent
- ▶ If confidence $\geq$ threshold: auto-resolve and trigger Explainability Engine
- ▶ If confidence $<$ threshold: route to human review queue with full evidence summary pre-loaded
- ▶ Decision + explanation delivered to card member and merchant portals simultaneously
- ▶ All actions written to immutable audit log in real time

6. Implementation & Feasibility

6.1 Build Phases

Phase	Timeline	Deliverables
Phase 1 — Foundation	Weeks 1–3	Core dispute filing flow, transaction evidence collection, FairScore™ v1 (rule-based), card member and merchant portals (MVP)
Phase 2 — AI Layer	Weeks 4–6	NLP evidence parser, XGBoost FairScore™ model trained on synthetic dispute data, pgvector precedent engine
Phase 3 — Explainability	Weeks 7–8	LLM-powered explanation generation, SHAP integration, audit log with hash chaining
Phase 4 — Ops & Polish	Weeks 9–10	AMEX Ops dashboard, model monitoring, fairness metrics, end-to-end testing and optimisation

6.2 Why This is Technically Feasible

- ▶ **Off-the-shelf AI components:** spaCy, Hugging Face, XGBoost, and pgvector are mature, production-ready libraries — no research-level AI required
- ▶ **Modular architecture:** Each layer (evidence, scoring, explanation) is independently deployable and testable
- ▶ **Graceful degradation:** If evidence collection from a source fails, the system continues with available data — never blocks resolution

- ▶ **Human-in-the-loop by design:** Low-confidence cases route to human review automatically, so the system is safe to deploy before 100% accuracy is achieved
- ▶ **AMEX integration-ready:** Architecture is designed around standard financial APIs and webhook patterns that AMEX systems already support

6.3 Scalability

- ▶ Stateless FastAPI services scale horizontally on AWS Lambda with zero infrastructure changes
- ▶ pgvector handles millions of case embeddings with sub-100ms retrieval
- ▶ FairScore™ inference runs in < 200ms per case — parallelisable across thousands of concurrent disputes
- ▶ The audit log architecture supports petabyte-scale storage with immutability guarantees

7. Business Impact & Relevance

7.1 Quantified Impact

95% Reduction in resolution time	\$28M+ Annual ops savings (1M disputes)	40% Reduction in appeal rates	2× Merchant satisfaction uplift
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7.2 Strategic Relevance to AMEX

- ▶ **Operational cost reduction:** At ~\$40 cost per manually handled dispute and millions of cases annually, even a 70% automation rate translates to tens of millions in savings
- ▶ **Card member retention:** Dispute experience is a leading driver of card member churn — fast, fair resolution directly improves retention
- ▶ **Merchant network health:** AMEX differentiates on merchant relationships. A fair, transparent dispute process strengthens those relationships versus competitor networks
- ▶ **Regulatory compliance:** Immutable audit trails and explainable decisions future-proof AMEX against increasing regulatory scrutiny of automated financial decisions
- ▶ **Platform extensibility:** The FairScore™ and Explainability engines can be reused across other AMEX decisioning workflows beyond disputes

7.3 Success Metrics

Metric	Target
Dispute Resolution Time	Baseline: 14–45 days → Target: < 5 minutes for auto-resolvable cases
Auto-Resolution Rate	Target: ≥ 70% of inbound disputes resolved without human intervention
Appeal Rate	Target: ≥ 40% reduction versus current baseline (driven by explainability)
Card Member Satisfaction (CSAT)	Target: +15 point improvement on dispute experience CSAT score

Metric	Target
Merchant Satisfaction	Target: ≥ 30% improvement on dispute transparency satisfaction survey
Audit Completeness	Target: 100% of decisions have a complete, immutable audit record
Model Fairness (Gini coefficient)	Decision rate parity within ±5% across card member demographic segments

8. Conclusion

Dispute resolution is not a back-office problem — it is a front-line customer experience and a direct driver of card member and merchant loyalty. Today's process is slow, opaque, and manual. ResolveIQ changes that.

By combining zero-touch evidence collection, a dual-perspective fairness model, and plain-English explainability, ResolveIQ delivers resolutions in minutes instead of weeks — with the transparency and auditability that both parties and regulators require.

ResolveIQ is not just a faster dispute tool. It is the foundation of a trust layer between card members, merchants, and the issuer — built on AI that is fair, explainable, and auditable by design.

We are confident that ResolveIQ addresses the challenge comprehensively, is technically feasible within the proposed timeline, and delivers measurable, quantifiable business value for AMEX and its stakeholders.